

CLASS 9 PHYSICS NCERT

CHAPTER NAME : MOTION

Q1. An object has moved through a distance. Can it have zero displacement? If yes, support your answer with an example.

Answer:

Yes, an object can have zero displacement even after moving a certain distance.

Distance is the total path travelled by an object, whereas displacement is the shortest distance between the initial and final positions.

If the initial and final positions are the same, the displacement is zero.

Example:

If a person walks 10 m forward and then 10 m backward, the total distance travelled is 20 m, but the displacement is 0 m because the person returns to the starting point.

Important Point:

Displacement can be zero even when distance is not zero.

Q2. A farmer moves along the boundary of a square field of side 10 m in 40 s. What will be the magnitude of displacement at the end of 2 minutes 20 seconds from his initial position?

Answer:

Given:

Side of square = 10 m

Time taken to complete one round = 40 s

Total time = 2 minutes 20 seconds = 140 s

Number of rounds completed = $140 / 40 = 3.5$ rounds

After 3 complete rounds, the farmer returns to the starting point.
After an additional half round, the farmer reaches the diagonally opposite corner of the square.

Displacement is equal to the diagonal of the square:

$$\text{Displacement} = \sqrt{(10^2 + 10^2)}$$

$$\text{Displacement} = \sqrt{200}$$

$$\text{Displacement} = 10\sqrt{2} \text{ m} \approx 14.14 \text{ m}$$

Final Answer:

$$\text{Displacement} = 10\sqrt{2} \text{ m or approximately } 14.14 \text{ m}$$

Important Point:

Displacement depends only on the initial and final positions, not on the path taken.

Q3. Which of the following is true for displacement?

- (a) It cannot be zero.
- (b) Its magnitude is greater than the distance travelled by the object.

Answer:

Both statements are incorrect.

(a) is false because displacement can be zero when the initial and final positions are the same.

(b) is false because displacement is always less than or equal to the distance travelled. It can never be greater than distance.

Correct Concept:

Displacement can be zero and is always less than or equal to distance.

Key Points for Revision:

- Distance is always greater than or equal to displacement.
- Displacement can be zero.

- Displacement is the shortest path between two points.
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Q4. An athlete completes one round of a circular track of diameter 200 m in 40 s. What will be the distance covered and the displacement at the end of 2 minutes 20 s?

Answer:

Given:

Diameter of circular track = 200 m

Time for one round = 40 s

Total time = 2 minutes 20 seconds = 140 s

Number of rounds = $140 / 40 = 3.5$ rounds

Distance covered in one round = circumference of circle

$= \pi \times \text{diameter} = \pi \times 200 = 200\pi \text{ m}$

Total distance = $3.5 \times 200\pi = 700\pi \text{ m} \approx 2199 \text{ m}$

After 3.5 rounds, the athlete reaches the point diametrically opposite to the starting point.

Displacement = diameter of circle = 200 m

Final Answer:

Distance $\approx 2199 \text{ m}$

Displacement = 200 m

Q5. Joseph jogs from A to B (300 m) in 2 min 30 s and then back 100 m to C in 1 min. Find average speed and velocity.

Answer:

(a) From A to B

Distance = 300 m

Time = 2 min 30 s = 150 s

Average speed = Distance / Time = $300 / 150 = 2$ m/s

Displacement = 300 m

Average velocity = $300 / 150 = 2$ m/s

(b) From A to C

Total distance = $300 + 100 = 400$ m

Total time = $150 + 60 = 210$ s

Average speed = $400 / 210 \approx 1.90$ m/s

Displacement from A to C = $300 - 100 = 200$ m

Average velocity = $200 / 210 \approx 0.95$ m/s

Q6. Abdul's average speed during a round trip

Answer:

Let total distance = d

Time taken to go = $d / 20$

Time taken to return = $d / 30$

Total time = $d/20 + d/30$

= $(3d + 2d)/60 = 5d/60 = d/12$

Total distance = $2d$

Average speed = Total distance / Total time

= $2d / (d/12) = 24$ km/h

Final Answer:

Average speed = 24 km/h

Q7. A motorboat accelerates at 3.0 m/s^2 for 8 s from rest. Find distance travelled.

Answer:

Given:

Initial velocity (u) = 0

Acceleration (a) = 3.0 m/s^2

Time (t) = 8 s

Using formula:

$$s = ut + (1/2)at^2$$

$$s = 0 + (1/2 \times 3 \times 8^2)$$

$$s = (1/2 \times 3 \times 64)$$

$$s = 96 \text{ m}$$

Final Answer:

Distance travelled = 96 m

Q8. Graph-based Question

(a) Distance from graph:

The distance travelled is represented by the area under the speed-time graph during the given time interval. Shade the region under the graph.

(b) Uniform motion:

The part of the graph which is a horizontal straight line represents uniform motion (constant speed).

Key Points for Revision:

- Distance = total path length
- Displacement = shortest distance between two points
- Average speed = total distance / total time
- Average velocity = displacement / total time

- Area under speed-time graph = distance
- Horizontal line in graph = uniform motion

Q9. State which of the following situations are possible and give an example for each:

(a) An object with a constant acceleration but with zero velocity

Answer:

Yes, this situation is possible.

An object can have zero velocity at a particular instant but still have constant acceleration.

Example:

When a ball is thrown vertically upwards, at the highest point its velocity becomes zero, but it still has acceleration due to gravity (9.8 m/s^2 downward).

(b) An object moving with an acceleration but with uniform speed

Answer:

Yes, this is possible.

Acceleration occurs when there is a change in velocity. Velocity changes if either speed or direction changes.

Even if speed is constant, a change in direction results in acceleration.

Example:

An object moving in a circular path with constant speed (uniform circular motion), like a satellite orbiting Earth.

(c) An object moving in a certain direction with an acceleration in the perpendicular direction

Answer:

Yes, this is possible.

When acceleration acts perpendicular to the direction of motion, it changes only the direction, not the speed.

Example:

In circular motion, the centripetal acceleration always acts towards the center, perpendicular to the direction of motion.

Q10. An artificial satellite is moving in a circular orbit of radius 42250 km. Calculate its speed if it takes 24 hours to revolve around the Earth.

Answer:

Given:

Radius of orbit (r) = 42250 km

Time period (T) = 24 hours = 86400 s

Distance covered in one revolution = circumference of circle

$$= 2\pi r = 2 \times \pi \times 42250 \text{ km}$$

$$= 84500\pi \text{ km}$$

Speed = Distance / Time

$$= (84500\pi \text{ km}) / (86400 \text{ s})$$

$$= (84500 \times 3.14) / 86400$$

$$\approx 265330 / 86400$$

$$\approx 3.07 \text{ km/s}$$

Final Answer:

Speed of satellite $\approx 3.07 \text{ km/s}$

Key Points for Revision:

- **Velocity can be zero even when acceleration is not zero.**

- Acceleration can occur due to change in direction.
 - In circular motion, acceleration is always perpendicular to velocity.
 - Speed = distance $\div$ time
 - Circular motion distance = $2\pi r$
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Q11. When will you say a body is in (i) uniform acceleration? (ii) non-uniform acceleration?

Answer:

(i) Uniform Acceleration:

A body is said to have uniform acceleration when its velocity changes by equal amounts in equal intervals of time.

In this case, acceleration remains constant throughout the motion.

Example:

A freely falling object under gravity.

(ii) Non-uniform Acceleration:

A body is said to have non-uniform acceleration when its velocity changes by unequal amounts in equal intervals of time.

In this case, acceleration is not constant.

Example:

A car moving in a crowded street (speed changes irregularly).

Q12. A bus decreases its speed from 80 km h^{-1} to 60 km h^{-1} in 5 s. Find the acceleration of the bus.

Answer:

Given:

Initial velocity (u) = 80 km/h = $80 \times (5/18) = 22.22 \text{ m/s}$

Final velocity (v) = 60 km/h = $60 \times (5/18) = 16.67 \text{ m/s}$

Time (t) = 5 s

Using formula:

Acceleration (a) = $(v - u) / t$

$$a = (16.67 - 22.22) / 5$$

$$a = (-5.55) / 5$$

$$a = -1.11 \text{ m/s}^2$$

Final Answer:

Acceleration = -1.11 m/s^2

(Note: Negative sign indicates deceleration.)

Q13. A train starting from a railway station and moving with uniform acceleration attains a speed of 40 km h^{-1} in 10 minutes. Find its acceleration.

Answer:

Given:

Initial velocity (u) = 0 (starting from rest)

Final velocity (v) = 40 km/h = $40 \times (5/18) = 11.11 \text{ m/s}$

Time (t) = 10 minutes = 600 s

Using formula:

Acceleration (a) = $(v - u) / t$

$$a = (11.11 - 0) / 600$$

$$a = 11.11 / 600$$

$$a \approx 0.0185 \text{ m/s}^2$$

Final Answer:

Acceleration $\approx 0.0185 \text{ m/s}^2$

Key Points for Revision:

- **Uniform acceleration → constant acceleration**
 - **Non-uniform acceleration → changing acceleration**
 - **$a = (v - u) / t$**
 - **Convert km/h to m/s using ($\times 5/18$)**
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CHAPTER : FORCE AND LAW OF MOTION

A1. Distance–Time Table Analysis

Given Data:

Time (s): 0, 1, 2, 3, 4, 5, 6, 7

Distance (m): 0, 1, 8, 27, 64, 125, 216, 343

(a) What conclusion can you draw about the acceleration?

Answer:

The distances follow the pattern:

$0^3, 1^3, 2^3, 3^3, 4^3, 5^3, 6^3, 7^3$

This means:

Distance $\propto$ (time)³

Since distance is not proportional to time or time², the acceleration is **not constant**.

Conclusion:

Acceleration is **increasing (non-uniform acceleration)**.

(b) What do you infer about the forces acting on the object?

Answer:

Since acceleration is increasing, the force acting on the object is also increasing (because Force $\propto$ Acceleration).

Conclusion:

The object is under the influence of a **non-uniform (increasing) force**.

A2. Force applied by each person**Answer:**

Given:

Mass (m) = 1200 kg

Acceleration (a) = 0.2 m/s^2

Using Newton's Second Law:

Force (F) = $m \times a = 1200 \times 0.2 = 240 \text{ N}$

This force is applied by **3 persons**, so force by each person:

Force per person = $240 / 3 = 80 \text{ N}$

Final Answer:

Each person applies a force of **80 N**

A3. Force of the nail on the hammer**Answer:**

Given:

Mass (m) = 500 g = 0.5 kg

Initial velocity (u) = 50 m/s

Final velocity (v) = 0

Time (t) = 0.01 s

Acceleration (a) = $(v - u) / t$

= $(0 - 50) / 0.01$

= -5000 m/s^2

$$\begin{aligned}\text{Force (F)} &= m \times a \\ &= 0.5 \times (-5000) \\ &= -2500 \text{ N}\end{aligned}$$

Final Answer:

Force = **2500 N** (opposite in direction to motion)

A4. Acceleration, change in momentum, and force

Answer:

Given:

Mass (m) = 1200 kg

Initial velocity (u) = 90 km/h = 25 m/s

Final velocity (v) = 18 km/h = 5 m/s

Time (t) = 4 s

(i) Acceleration:

$$\begin{aligned}a &= (v - u) / t \\ &= (5 - 25) / 4 \\ &= -20 / 4 \\ &= -5 \text{ m/s}^2\end{aligned}$$

(ii) Change in momentum:

Initial momentum = $m \times u = 1200 \times 25 = 30000 \text{ kg}\cdot\text{m/s}$

Final momentum = $m \times v = 1200 \times 5 = 6000 \text{ kg}\cdot\text{m/s}$

Change in momentum = $6000 - 30000 = -24000 \text{ kg}\cdot\text{m/s}$

(iii) Force:

$$F = m \times a = 1200 \times (-5) = -6000 \text{ N}$$

Final Answers:

- Acceleration = **-5 m/s^2**
 - Change in momentum = **$-24000 \text{ kg}\cdot\text{m/s}$**
 - Force = **6000 N (opposite direction)**
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Key Points for Revision:

- If distance $\propto t^3 \rightarrow$ acceleration is increasing
 - $F = m \times a$
 - $a = (v - u) / t$
 - Momentum = mass $\times$ velocity
 - Force = rate of change of momentum
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Q1. Can an object have non-zero velocity when net external force is zero?

Answer:

Yes, it is possible.

According to Newton's First Law, if the net external force on an object is zero, it will continue to move with constant velocity.

Condition:

The object must be moving with constant velocity (uniform motion) in a straight line.

Q2. Why does dust come out when a carpet is beaten?

Answer:

When the carpet is beaten, it moves suddenly, but the dust particles tend to remain at rest due to inertia. Hence, they get separated and fall off.

Q3. Why should luggage on a bus roof be tied with a rope?

Answer:

Due to inertia, luggage may continue in its state of motion when the bus suddenly stops or starts. Tying it with a rope prevents it from falling.

Q4. Correct option for ball coming to rest

Answer:

Correct option: (c) there is a force on the ball opposing the motion.

Q5. Truck rolling down a hill

Answer:

Given:

Distance (s) = 400 m

Time (t) = 20 s

Initial velocity (u) = 0

Mass = 7 tonnes = 7000 kg

Using:

$$s = ut + (1/2)at^2$$

$$400 = 0 + (1/2)a(20^2)$$

$$400 = 200a$$

$$a = 2 \text{ m/s}^2$$

Force:

$$F = m \times a = 7000 \times 2 = 14000 \text{ N}$$

Final Answer:

$$\text{Acceleration} = 2 \text{ m/s}^2$$

$$\text{Force} = 14000 \text{ N}$$

Q6. Force of friction on stone

Answer:

Given:

$$\text{Mass} = 1 \text{ kg}$$

$$u = 20 \text{ m/s}, v = 0, s = 50 \text{ m}$$

Using:

$$v^2 = u^2 + 2as$$

$$0 = 400 + 2a(50)$$

$$0 = 400 + 100a$$

$$a = -4 \text{ m/s}^2$$

$$\text{Friction force} = m \times a = 1 \times (-4) = -4 \text{ N}$$

Final Answer:

$$\text{Friction force} = 4 \text{ N (opposite direction)}$$

Q7. Engine pulling train

Answer:

$$\text{Total mass} = 8000 + (5 \times 2000) = 18000 \text{ kg}$$

$$(a) \text{ Net force} = 40000 - 5000 = 35000 \text{ N}$$

(b) Acceleration:

$$a = F/m = 35000 / 18000 \approx 1.94 \text{ m/s}^2$$

Q8. Force to stop vehicle

Answer:

Given:

Mass = 1500 kg

Acceleration = -1.7 m/s^2

$$F = m \times a = 1500 \times (-1.7) = -2550 \text{ N}$$

Final Answer:

Force = 2550 N (opposite direction)

Q9. Momentum formula

Answer:

Correct option: (d) mv

Q10. Friction force at constant velocity

Answer:

At constant velocity, net force = 0

So, friction force = applied force = 200 N

Q11. Why truck does not move?

Answer:

The student's logic is incorrect.

Action and reaction forces act on different bodies, so they do not cancel each other.

The truck does not move because the applied force is not enough to overcome friction.

Q12. Change in momentum (hockey ball)

Answer:

Mass = 200 g = 0.2 kg

Initial velocity = 10 m/s

Final velocity = -5 m/s

Change in momentum:

$$\Delta p = m(v - u)$$

$$= 0.2(-5 - 10)$$

$$= 0.2(-15) = -3 \text{ kg}\cdot\text{m/s}$$

Magnitude = 3 kg·m/s

Q13. Bullet penetration

Answer:

Mass = 10 g = 0.01 kg

u = 150 m/s, v = 0, t = 0.03 s

Acceleration:

$$a = (v - u)/t = -150/0.03 = -5000 \text{ m/s}^2$$

Distance:

$$s = ut + (1/2)at^2$$

$$= 150(0.03) + (1/2)(-5000)(0.03^2)$$

$$= 4.5 - 2.25 = 2.25 \text{ m}$$

Force:

$$F = m \times a = 0.01 \times (-5000) = -50 \text{ N}$$

Final Answer:

Distance = 2.25 m

Force = 50 N

Q14. Collision and momentum

Answer:

Initial momentum:

$$= 1 \times 10 = 10 \text{ kg}\cdot\text{m/s}$$

Final velocity:

$$v = (10) / (1 + 5) = 10/6 \approx 1.67 \text{ m/s}$$

Final momentum:

$$= 6 \times 1.67 \approx 10 \text{ kg}\cdot\text{m/s}$$

Q15. Momentum and force

Answer:

Mass = 100 kg

Initial momentum = $100 \times 5 = 500 \text{ kg}\cdot\text{m/s}$

Final momentum = $100 \times 8 = 800 \text{ kg}\cdot\text{m/s}$

Force:

$$a = (8 - 5)/6 = 0.5 \text{ m/s}^2$$

$$F = 100 \times 0.5 = 50 \text{ N}$$

Q16. Insect and car explanation

Answer:

Rahul is correct.

According to Newton's Third Law, both experience equal and opposite forces and equal change in momentum.

The insect dies because its mass is very small, so acceleration is very large.

Q17. Momentum of falling dumb-bell

Answer:

Mass = 10 kg

Height = 0.8 m

Velocity:

$$v^2 = 2gh = 2 \times 10 \times 0.8 = 16$$

$$v = 4 \text{ m/s}$$

Momentum:

$$p = m \times v = 10 \times 4 = 40 \text{ kg}\cdot\text{m/s}$$

Key Points for Revision:

- **Momentum = mass $\times$ velocity**
 - **Force = rate of change of momentum**
 - **Newton's laws explain motion clearly**
 - **Friction opposes motion**
 - **Action $\neq$ cancellation (acts on different bodies)**
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Q18. Which of the following has more inertia?

Answer:

Inertia depends on the mass of the object. Greater the mass, greater the inertia.

(a) A stone has more inertia than a rubber ball (same size but more mass).

(b) A train has more inertia than a bicycle (very large mass).

(c) A five-rupee coin has more inertia than a one-rupee coin (greater mass).

Q19. Identify the number of times velocity changes and the agent applying force

Answer:

Velocity changes whenever there is a change in speed or direction.

Total number of changes in velocity = 3 times

- 1. First player kicks the football → velocity changes**
Agent: First player
- 2. Second player kicks the football towards the goal → velocity changes**
Agent: Second player
- 3. Goalkeeper stops and kicks the football → velocity changes**
Agent: Goalkeeper

Q20. Why do leaves fall when a branch is shaken?

Answer:

When a branch is shaken, it moves suddenly, but the leaves tend to remain at rest due to inertia of rest.

This causes the leaves to detach and fall.

Q21. Why do we fall forward or backward in a moving bus?

Answer:

This happens due to inertia.

- When a moving bus suddenly stops, the lower part of the body stops but the upper part continues moving forward due to inertia → we fall forward.**
- When a bus starts suddenly, the lower part moves forward but the upper part remains at rest due to inertia → we fall backward.**

Key Points for Revision:

- **Inertia depends on mass**
 - **Velocity changes when speed or direction changes**
 - **Inertia of rest and motion explain daily life situations**
 - **Sudden motion causes imbalance due to inertia**
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Class 9 PHYSICS Gravitation (NCERT Solutions)

Q1. How does gravitational force change when distance is reduced to half?

Answer:

According to the universal law of gravitation:

$$\text{Force} \propto 1 / (\text{distance})^2$$

If distance is reduced to half:

$$\text{New force} = 1 / ((d/2)^2) = 4 / d^2$$

Conclusion:

Gravitational force becomes 4 times.

Q2. Why does a heavy object not fall faster than a light object?

Answer:

Although gravitational force depends on mass, acceleration due to gravity (g) is independent of mass.

Thus, all objects fall with the same acceleration in the absence of air resistance.

Q3. Gravitational force between Earth and 1 kg object

Answer:

Given:

$$G = 6.7 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$$

$$\text{Mass of Earth (M)} = 6 \times 10^{24} \text{ kg}$$

$$\text{Mass of object (m)} = 1 \text{ kg}$$

$$\text{Radius of Earth (r)} = 6.4 \times 10^6 \text{ m}$$

Using formula:

$$F = GMm / r^2$$

$$F = (6.7 \times 10^{-11} \times 6 \times 10^{24} \times 1) / (6.4 \times 10^6)^2$$

$$F \approx 9.8 \text{ N}$$

Final Answer:

Force $\approx 9.8 \text{ N}$

Q4. Force between Earth and Moon

Answer:

The Earth attracts the Moon with the same force with which the Moon attracts the Earth.

This is according to Newton's Third Law of Motion (equal and opposite forces).

Q5. Why does Earth not move towards the Moon?

Answer:

Earth also moves slightly, but its motion is not noticeable because its mass is very large compared to the Moon.

Q6. Effect on gravitational force

Answer:

(i) If mass is doubled $\rightarrow$ Force becomes 2 times

(ii) If distance is doubled $\rightarrow$ Force becomes 1/4

If distance is tripled $\rightarrow$ Force becomes 1/9

(iii) If both masses are doubled $\rightarrow$ Force becomes 4 times

Q7. Importance of universal law of gravitation

Answer:

- Explains force binding us to Earth
 - Explains motion of planets around the Sun
 - Explains tides due to Moon and Sun
 - Explains motion of artificial satellites
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Q8. What is acceleration of free fall?

Answer:

Acceleration of free fall is the acceleration with which an object falls towards Earth due to gravity.

Its value is $g = 9.8 \text{ m/s}^2$.

Q9. What is gravitational force between Earth and an object called?

Answer:

It is called weight of the object.

Q10. Gold weight at poles vs equator

Answer:

No, the friend will not agree.

Weight depends on g . Since g is greater at poles than at equator, the gold will weigh less at the equator.

Q11. Why does a sheet fall slower than a crumpled ball?

Answer:

A sheet experiences more air resistance due to larger surface area, so it falls slower than a crumpled ball.

Q12. Weight on Earth and Moon

Answer:

Mass = 10 kg

Weight on Earth:

$$W = mg = 10 \times 9.8 = 98 \text{ N}$$

Weight on Moon:

$$g (\text{moon}) = g/6 = 9.8/6 \approx 1.63$$

$$W = 10 \times 1.63 \approx 16.3 \text{ N}$$

Key Points for Revision:

- $F \propto 1/r^2$
- $g = 9.8 \text{ m/s}^2$
- $\text{Weight} = mg$
- Gravity acts equally on both bodies
- Air resistance affects falling speed

Q13. A ball is thrown vertically upwards with a velocity of 49 m/s.

Answer:

Given:

Initial velocity (u) = 49 m/s

Final velocity (v) = 0 (at maximum height)

$$g = 9.8 \text{ m/s}^2$$

(i) Maximum height:

Using:

$$v^2 = u^2 - 2gh$$

$$0 = (49)^2 - 2 \times 9.8 \times h$$

$$h = 2401 / 19.6$$

$$h = 122.5 \text{ m}$$

(ii) Total time:

Using:

$$v = u - gt$$

$$0 = 49 - 9.8t$$

$$t = 5 \text{ s (time to reach top)}$$

$$\text{Total time} = 2 \times 5 = 10 \text{ s}$$

Q14. Stone falling from height 19.6 m

Answer:

Given:

$$u = 0, h = 19.6 \text{ m}, g = 9.8 \text{ m/s}^2$$

Using:

$$v^2 = u^2 + 2gh$$

$$v^2 = 2 \times 9.8 \times 19.6 = 384.16$$

$$v = 19.6 \text{ m/s}$$

Q15. Stone thrown upward ($u = 40 \text{ m/s}$, $g = 10 \text{ m/s}^2$)

Answer:

Maximum height:

$$0 = 40^2 - 2 \times 10 \times h$$

$$h = 1600 / 20 = 80 \text{ m}$$

Net displacement:

$$= 0 \text{ (returns to starting point)}$$

Total distance:
= 80 + 80 = 160 m

Q16. Force between Earth and Sun

Answer:

Given:

$$G = 6.7 \times 10^{-11}$$

$$M_1 = 6 \times 10^{24} \text{ kg}$$

$$M_2 = 2 \times 10^{30} \text{ kg}$$

$$r = 1.5 \times 10^{11} \text{ m}$$

Using:

$$F = GM_1M_2 / r^2$$

$$F \approx 3.6 \times 10^{22} \text{ N}$$

Q17. Two stones meeting

Answer:

Let time = t

For falling stone:

$$s_1 = (1/2)gt^2 = 5t^2$$

For upward stone:

$$s_2 = ut - (1/2)gt^2 = 25t - 5t^2$$

Total height = 100 m

$$5t^2 + (25t - 5t^2) = 100$$

$$25t = 100$$

$$t = 4 \text{ s}$$

Position:

$$s_1 = 5 \times (4^2) = 80 \text{ m from top}$$

$$= 20 \text{ m above ground}$$

Q18. Ball returns in 6 s

Answer:

Total time = 6 s → time to reach top = 3 s

(a) Initial velocity:

$$v = u - gt$$

$$0 = u - 9.8 \times 3$$

$$u = 29.4 \text{ m/s}$$

(b) Maximum height:

$$h = ut - (1/2)gt^2$$

$$= 29.4 \times 3 - 4.9 \times 9$$

$$= 88.2 - 44.1 = 44.1 \text{ m}$$

(c) Position after 4 s:

$$s = ut - (1/2)gt^2$$

$$= 29.4 \times 4 - 4.9 \times 16$$

$$= 117.6 - 78.4 = 39.2 \text{ m above ground}$$

Q19. Direction of buoyant force

Answer:

Buoyant force always acts in the upward direction.

Q20. Why plastic block rises in water?

Answer:

Because buoyant force acting on it is greater than its weight, so it rises to the surface.

Q21. Will substance float or sink?

Answer:

Density = mass / volume = $50 / 20 = 2.5 \text{ g/cm}^3$

Since density > water (1 g/cm^3), it will sink.

Q22. Floating of sealed packet

Answer:

Density = $500 / 350 \approx 1.43 \text{ g/cm}^3$

Since density > water $\rightarrow$ it will sink

Mass of water displaced = 500 g (equal to mass of object)

Key Points for Revision:

- **$v^2 = u^2 \pm 2as$**
 - **$s = ut + (1/2)at^2$**
 - **Buoyant force acts upward**
 - **Object floats if density < liquid**
 - **Object sinks if density > liquid**
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Q23. State the universal law of gravitation.

Answer:

The universal law of gravitation states that:

“Every object in the universe attracts every other object with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres.”

Q24. Write the formula for gravitational force between Earth and an object.

Answer:

$$F = \frac{G M m}{r^2}$$

Where:

F = gravitational force

G = universal gravitational constant

M = mass of Earth

m = mass of object

r = distance between centres (radius of Earth)

Q25. What do you mean by free fall?

Answer:

Free fall is the motion of an object when it falls towards the Earth under the influence of gravity alone, without any other force acting on it.

Q26. What do you mean by acceleration due to gravity?

Answer:

Acceleration due to gravity is the acceleration gained by an object due to the gravitational force of the Earth.

Its value is approximately 9.8 m/s^2 and is denoted by g .

Q27. Difference between mass and weight

Answer:

Mass	Weight
Amount of matter in an object	Force of gravity acting on object
SI unit: kilogram (kg)	SI unit: newton (N)
Constant everywhere	Varies with location
Scalar quantity	Vector quantity
Measured using balance	Measured using spring balance

Q28. Why is weight on the Moon 1/6th of that on Earth?

Answer:

Weight depends on gravitational acceleration (g).

The Moon's gravitational pull is weaker than Earth's because:

- **It has less mass**
- **Smaller radius**

Thus, gravity on Moon $\approx 1/6$ of Earth

So, weight on Moon = $(1/6) \times$ weight on Earth

Key Points for Revision:

- **$F = GMm / r^2$**
- **$g = 9.8 \text{ m/s}^2$**

- **Weight = mg**
 - **Mass is constant, weight varies**
 - **Free fall = motion under gravity only**
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